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EXPERIMENT-8

QUESTION 1

Write a menu-driven C program to input two numbers and swap them using a third variable & without using third variable. Also display the values before and after swapping.

1. Problem Understanding

Input: Two numbers `a` and `b`

Menu options:

1. Swap using third variable
2. Swap without using third variable
3. Exit

Output: Values of `a` and `b` before and after swapping.

Logic:

Using third variable:

- `temp = a`
- `a = b`
- `b = temp`

Without third variable:

- `a = a + b`
- `b = a - b`
- `a = a - b`

2. Test Case Table

Test Case	Input	Expected Output	Actual Output	Results
1	a = 10, b = 20, Choice = 1	Before: a = 10, b = 20; After: a = 20, b = 10	Same as expected	Pass

2	a = 5, b = 15, Choice = 2	Before: a = 5, b = 15; After: a = 15, b = 5	Same as expected	Pass
3	a = 8, b = 12, Choice = 3	Program exits	Same as expected	Pass

3. Algorithm

1. Start.
2. Input two numbers `a` and `b`.
3. Display the values before swapping.
4. Display the menu:

    - Swap using third variable
    - Swap without third variable
    - Exit

5. Input the user's choice.
6. If choice is 1:

- Store a in temp.
- Store b in a.
- Store temp in b.
- Display the swapped values.

7. If choice is 2:

- Add `a` and `b` and store the result in `a`.
- Subtract `b` from `a` and store it in `b`.
- Subtract `b` from `a` and store it in `a`.
- Display the swapped values.

- If choice is 3, exit the program.
- If any other choice is entered, display "Invalid choice".
- Stop.

4.Pseudocode

START

INPUT a, b

DISPLAY "Before swapping", a, b

DISPLAY "1. Swap using third variable"

DISPLAY "2. Swap without third variable"

```
DISPLAY "3. Exit"
INPUT choice
IF choice = 1 THEN
    temp = a
    a = b
    b = temp
    DISPLAY a, b
ELSE IF choice = 2 THEN
    a = a + b
    b = a - b
    a = a - b
    DISPLAY a, b
ELSE IF choice = 3 THEN
    DISPLAY "Exit"
ELSE
    DISPLAY "Invalid choice"
STOP
```

5. FLOWCHART

Start

Input two numbers a and b

Display values before swapping

Display Menu

1. Swap using third variable

2. Swap without third variable

3. Exit

Enter choice

Check choice

If choice = 1

temp = a

a = b

b = temp

Display values after swapping

Stop

If choice = 2

a = a + b

b = a - b

a = a - b

Display values after swapping

Stop

If choice = 3

Exit

Stop

6.FLOWCHART

```
 #include <stdio.h>
int main()
{
    int a, b, temp, choice;
    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);
    printf("\nBefore swapping: a = %d, b = %d\n", a, b);
    printf("\nMenu");
    printf("\n1. Swap using third variable");
    printf("\n2. Swap without third variable");
    printf("\n3. Exit");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);
    switch(choice)
    {
        case 1:
            temp = a;
            a = b;
            b = temp;
            printf("After swapping: a = %d, b = %d\n", a, b);
            break;
```

```
        case 2:
            a = a + b;
            b = a - b;
            a = a - b;
            printf("After swapping: a = %d, b = %d\n", a, b);
            break;
        case 3:
            printf("Exiting program\n");
            break;
        default:
            printf("Invalid choice\n");
    }
    return 0;
}
```

7.Compilation And Execution

```
Enter your choice: 2
After swapping: a = 15, b = 5

┌──(kali㉿kali)-[~]
└─$ gcc hdoc9.c -o hdoc9

┌──(kali㉿kali)-[~]
└─$ ./hdoc9
Enter two numbers: 8 12

Before swapping: a = 8, b = 12

Menu
1. Swap using third variable
2. Swap without third variable
3. Exit
Enter your choice: 3
Exiting program
```

QUESTION 2

Write a menu-driven C program to find the sum of the first n natural numbers and sum of squares of the first n natural numbers.

1. Problem Understanding

Input: Value of n

Menu:

1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
3. Exit

Formulas:

Sum of first n natural numbers:

Sum = n * (n + 1) / 2

Sum of squares:

Sum of squares = n * (n + 1) * (2*n + 1) / 6

2. Test Case

Test Case	Inputs	Expected Output	Actual Output	Results
1	n = 5, Choice = 1	Sum = 15	Sum = 15	Pass
2	n = 5, Choice = 2	Sum of squares = 55	Sum of squares = 55	Pass
3	n = 10, Choice =1	Sum = 55	Sum = 55	Pass

4	n = 10, Choice = 2	Sum of squares = 385	Sum of squares = 385	Pass

3. Algorithm

1. Start.
2. Input the value of `n`.
3. Display the menu.
4. Input the user's choice.
5. If choice is 1:
   - Calculate `sum = n * (n + 1) / 2`.
   - Display the sum.
6. If choice is 2:
   - Calculate `sumsq = n * (n + 1) * (2*n + 1) / 6`.
   - Display the sum of squares.
7. If choice is 3, exit.
8. Otherwise display "Invalid choice".
9. Stop.

4. Pseudocode

```
START
INPUT n
DISPLAY "1. Sum of first n natural numbers"
DISPLAY "2. Sum of squares of first n natural numbers"
DISPLAY "3. Exit"
INPUT choice
IF choice = 1 THEN
   sum = n * (n + 1) / 2
   DISPLAY sum
ELSE IF choice = 2 THEN
   sumsq = n * (n + 1) * (2*n + 1) / 6
   DISPLAY sumsq
ELSE IF choice = 3 THEN
   DISPLAY "Exit"
ELSE
   DISPLAY "Invalid choice"
STOP
```

5. Flowchart

Start
Input n
Display Menu

1. Sum of natural numbers
2. Sum of squares

3. Exit
   Input choice
   sum = n*(n+1)/2

Choice1:

Display sum

Choice2:

sumsq = n*(n+1)*(2*n+1)/6

Display sumsq

Choice3:

Stop

6. C Program

```
#include <stdio.h>

int main()
{
   int n, choice;
   int sum, sumsq;
   printf("Enter the value of n: ");
   scanf("%d", &n);
   printf("\nMenu");
   printf("\n1. Sum of first n natural numbers");
   printf("\n2. Sum of squares of first n natural numbers");
   printf("\n3. Exit")
   printf("\nEnter your choice: ");
   scanf("%d", &choice);
   switch(choice)
   {
      case 1:
         sum = n * (n + 1) / 2;
         printf("Sum of first %d natural numbers = %d\n", n, sum);
         break;
      case 2:
         sumsq = n * (n + 1) * (2 * n + 1) / 6;
         printf("Sum of squares of first %d natural numbers = %d\n",n, sumsq);
         break;
      case 3:
         printf("Exiting program\n");
         break;
      default:
         printf("Invalid choice\n");
   }
   return 0;
}
```

7.Compilation and Execution

```
┌──(kali㉿kali)-[~]
└─$ gcc hdoc9ii.c -o hdoc9ii

┌──(kali㉿kali)-[~]
└─$ ./hdoc9ii
Enter the value of n: 5

Menu
1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
3. Exit
Enter your choice: 1
Sum of first 5 natural numbers = 15

┌──(kali㉿kali)-[~]
└─$ gcc hdoc9ii.c -o hdoc9ii

┌──(kali㉿kali)-[~]
└─$ ./hdoc9ii
Enter the value of n: 5

Menu
1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
3. Exit
Enter your choice: 2
```

```
Sum of squares of first 5 natural numbers = 55

┌──(kali㉿kali)-[~]
└─$ gcc hdoc9ii.c -o hdoc9ii

┌──(kali㉿kali)-[~]
└─$ ./hdoc9ii
Enter the value of n: 10

Menu
1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
3. Exit
Enter your choice: 1
Sum of first 10 natural numbers = 55

┌──(kali㉿kali)-[~]
└─$ gcc hdoc9ii.c -o hdoc9ii

┌──(kali㉿kali)-[~]
└─$ ./hdoc9ii
Enter the value of n: 10

Menu
1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
3. Exit
Enter your choice: 2
Sum of squares of first 10 natural numbers = 385
```

QUESTION 3

Write a menu-driven C program to calculate simple interest, compound amount, compound interest, and total payable amount for given principal, rate, and time.

1. Problem Understanding

Inputs:

- P = Principal amount
- R = Rate of interest
- T = Time in years

Simple Interest:

SI = (P * R * T) / 100

Compound Amount:

For annual compounding:

A = P * pow((1 + R/100), T)

Compound Interest:

CI = A - P

Total Payable Amount:

Total Payable Amount = A

2. Test Case

Test Case	Inputs	Expected Output	Actual Output	Results
1	P=1000, R=5, T=2, Choice=1	SI = 100.00	SI = 100.00	Pass
2	P=1000, R=5, T=2, Choice=2	Compound Amount = 1102.50	1102.50	Pass
3	P=1000, R=5, T=2, Choice=3	CI = 102.50	102.50	Pass
4	P=1000, R=5, T=2, Choice=4	Total Payable = 1102.50	1102.50	Pass

3. Algorithm

1. Start.
2. Input principal `P`, rate `R`, and time `T`.
3. Display the menu:
    - Simple Interest
    - Compound Amount
    - Compound Interest
    - Total Payable Amount
    - Exit
4. Input the user's choice.

5. If choice is 1:
    - Calculate SI = (P*R*T)/100.
    - Display simple interest.
6. If choice is 2:
    - Calculate compound amount.
    - Display compound amount.
7. If choice is 3:
    - Calculate compound amount.
    - Calculate `CI = A-P`.
    - Display compound interest.
8. If choice is 4:
    - Calculate compound amount.
    - Display it as total payable amount.
9. If choice is 5, exit.
10. Otherwise, display "Invalid choice".
11. Stop.

4. Pseudocode

```
START
INPUT P, R, T
DISPLAY "1. Simple Interest"
DISPLAY "2. Compound Amount"
DISPLAY "3. Compound Interest"
DISPLAY "4. Total Payable Amount"
DISPLAY "5. Exit"
INPUT choice
IF choice = 1 THEN
   SI = (P * R * T) / 100
   DISPLAY SI
ELSE IF choice = 2 THEN
   A = P * (1 + R/100)^T
   DISPLAY A
ELSE IF choice = 3 THEN
   A = P * (1 + R/100)^T
   CI = A - P
   DISPLAY CI
ELSE IF choice = 4 THEN
   A = P * (1 + R/100)^T
   DISPLAY A
ELSE IF choice = 5 THEN
   DISPLAY "Exit"
ELSE
   DISPLAY "Invalid choice"
```

STOP

5. Flowchart

Start
Input P, R, T
Display Menu

1. SI
2. Compound Amount
3. CI
4. Total Payable
5. Exit
   Input choice
   Choice?

Choice 1

SI = (P*R*T)/100

Display SI

Choice 2

A = P*pow(1+R/100,T)

Display A

Choice 3 → Rectangle:

A = P*pow(1+R/100,T)
CI = A-P

Display CI

Choice 4 A = P*pow(1+R/100,T)

Display Total Payable Amount

Choice 5

Stop

6. C Program

```
#include <stdio.h>
#include <math.h>
int main()
{
   double P, R, T, SI, A, CI;
```

```
    int choice;
    printf("Enter Principal: ");
    scanf("%lf", &P);
    printf("Enter Rate of Interest: ");
    scanf("%lf", &R);
    printf("Enter Time in years: ");
    scanf("%lf", &T);
    printf("\nMenu");
    printf("\n1. Simple Interest");
    printf("\n2. Compound Amount");
    printf("\n3. Compound Interest");
    printf("\n4. Total Payable Amount");
    printf("\n5. Exit");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);
    switch(choice)
    {
        case 1:
            SI = (P * R * T) / 100;
            printf("Simple Interest = %.2lf\n", SI);
            break;
        case 2:
            A = P * pow((1 + R / 100), T);
            printf("Compound Amount = %.2lf\n", A);
            break;
        case 3:
            A = P * pow((1 + R / 100), T);
            CI = A - P;
            printf("Compound Interest = %.2lf\n", CI);
            break;
        case 4:
            A = P * pow((1 + R / 100), T);
            printf("Total Payable Amount = %.2lf\n", A);
            break;
        case 5:
            printf("Exiting program\n");
            break;
        default:
            printf("Invalid choice\n");
    }
    return 0;
}
```

7.Compilation and Execution

```
┌──(kali㉿kali)-[~]
└─$ gcc hdoc9iii.c -o hdoc9iii -lm

┌──(kali㉿kali)-[~]
└─$ ./hdoc9iii
Enter Principal: 1000
Enter Rate of Interest: 5
Enter Time in years: 2

Menu
1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
5. Exit
Enter your choice: 1
Simple Interest = 100.00

┌──(kali㉿kali)-[~]
└─$ gcc hdoc9iii.c -o hdoc9iii -lm

┌──(kali㉿kali)-[~]
└─$ ./hdoc9iii
Enter Principal: 1000
Enter Rate of Interest: 5
Enter Time in years: 2

Menu
1. Simple Interest
```

```
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
5. Exit
Enter your choice: 2
Compound Amount = 1102.50

┌──(kali㉿kali)-[~]
└─$ gcc hdoc9iii.c -o hdoc9iii -lm

┌──(kali㉿kali)-[~]
└─$ ./hdoc9iii
Enter Principal: 1000
Enter Rate of Interest: 5
Enter Time in years: 2

Menu
1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
5. Exit
Enter your choice: 3
Compound Interest = 102.50

┌──(kali㉿kali)-[~]
└─$ gcc hdoc9iii.c -o hdoc9iii -lm
```

```
┌──(kali㉿kali)-[~]
└─$ ./hdoc9iii
Enter Principal: 1000
Enter Rate of Interest: 5
Enter Time in years: 2

Menu
1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount
5. Exit
Enter your choice: 4
Total Payable Amount = 1102.50
```